

Answer **all** questions.

1. (a) A student investigated the pH of several substances using universal indicator solution. The colour produced with each substance is shown.

Substance	Colour
hand wash	blue
battery fluid	red
water	green
lemon juice	orange
drain cleaner	purple

- (i) Name **one** substance in the table which is an alkali. [1]

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- (ii) State which substance is the strongest acid. [1]

.....

- (iii) Drain cleaner contains sodium hydroxide. [1]
Circle the correct formula of sodium hydroxide.

NaO NAOH NaOH NaOh



(b) The student carried out some reactions using hydrochloric acid. She recorded the following observations.

Reaction	Reactant added to acid	Observations
A	magnesium	fizzing temperature increase of 25 °C magnesium disappears
B	sodium hydroxide	no fizzing temperature increase of 8 °C
C	sodium carbonate	fizzing temperature increase of 5 °C
D	copper(II) oxide	no fizzing temperature increase of 11 °C mixture turns blue

(i) Reaction **A** produced a gas. Give the **letter** of the other reaction that produced a gas. Which observation shows that a gas was produced? [2]

Letter
Observation

(ii) One of the gases produced is hydrogen.
Describe how you could test for hydrogen gas. Give the observation you would expect for a positive test. [2]

Test
Observation

(iii) Give the **letter** of the reaction which was the **least** exothermic. Give the reason for your answer. [2]

Letter
Reason

